

Run/S&P/Unknown Classification Algorithm

Version: 0.51.0-Beta **Date:** 2026-03-04 **Category:** Algorithm Spec

--- Revision History ---

[0.51.0-Beta] - 2026-03-04

Changed

- Pass 1: Documented that run-detection uses a 10-minute + 1 second window (implementation detail).
- Pass 2: Clarified that the target QSO is excluded from both preceding and following window counts; only other contacts in the stream are counted.

[0.50.0-Beta] - 2026-02-02

Changed

- Synchronized Pass 1 and Pass 2 thresholds to 10 minutes and 3 QSOs to eliminate the misclassification of slow-rate stationary activity as S&P.
 - Updated "Pass 2" logic to ensure low-rate activity is correctly flagged as "Unknown."
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1. Purpose

The purpose of this algorithm is to analyze a contest log and infer the operator's activity for each contact, classifying it as **Run**, **Search & Pounce (S&P)**, or **Unknown**. Analysis is performed independently for each operating "stream" (a unique combination of MyCall, Band, and Mode).

2. Pass 1: Initial Classification (Sticky Run Logic)

The first pass uses a state machine to detect "Runs" based on frequency stability and activity rate.

- **Run Identification:** A Run is initiated when **3 QSOs** are detected on the same frequency within a **10-minute** window. (Implementation: the run-detection window is 10 minutes + 1 second for buffer and segment checks.)
- **Frequency Tolerance:** The algorithm allows for a tolerance of 0.1 kHz for CW or 0.5 kHz for Phone.

- **Retroactive Tagging:** When a Run is identified, the initial QSOs in the buffer that triggered the detection are retroactively reclassified from S&P to **Run**.
 - **Sticky State Maintenance:** Once active, the "Run" state is maintained for all subsequent QSOs on that frequency until:
 - **Time-Out:** 2 minutes pass without a QSO on the run frequency.
 - **Frequency Change:** 3 consecutive QSOs occur on a different frequency.
 - **Initial S&P:** Any QSO not meeting the above criteria is initially classified as **S&P**.
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3. Pass 2: Reclassification of Ambiguous Activity

The second pass refines the results by identifying periods where the activity rate is too low to reliably distinguish between a struggling Run and an S&P attempt.

- **Targeting S&P:** The algorithm re-examines only QSOs currently labeled as **S&P**.
 - **Synchronized Density Check:** For each target QSO, the algorithm counts **other** contacts in the same stream in two separate **10-minute** windows: one immediately preceding and one immediately following the contact (the target QSO itself is excluded from both counts).
 - **Classification to Unknown:** If the count in **both** windows is below the threshold of **3 QSOs**, the contact is reclassified as **Unknown**.
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4. Classification Matrix (Summary)

Activity Rate	Frequency Stability	Final Result
≥ 3 per 10 min	Static	Run
≥ 3 per 10 min	Agile	S&P
< 3 per 10 min	Any	Unknown